Stochastic Analysis

LectureNotes

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Chapter 1

Intro

Theorem 1.0.1 (Kunita-Watanabe)**.** Let $M \in \mathcal{M}_0^2$ and $H \in L^2(M)$. Then $\forall\ N \in \mathcal{M}_0^2$,

$$\mathbb{E}\left[(H \cdot M)_\infty \cdot N_\infty\right] = \mathbb{E}\left[(H \cdot [M,N])_\infty\right].$$

Moreover,

$$[H \cdot M, N] = H \cdot [M,N] \overset{def}{=\!=} \int H_s \,\mathrm{d}[M,N]_s.$$

Theorem 1.0.2. Let $M, N \in \mathcal{M}_0^2$. Then

$$\left(\int_0^\infty |H_s K_s| \,\mathrm{d}[M,N]_s\right)^2 \leq \int_0^\infty H_s^2 \,\mathrm{d}[M]_s \cdot \int_0^\infty K_s^2 \,\mathrm{d}[N]_s. \quad \text{a.s.}$$

Proof. $|H_s K_s| = H_s K_s \cdot \operatorname{sgn}(H_s K_s)$.
$|\mathrm{d}[M,N]_s| = \mathrm{d}[M,N]_s \cdot \operatorname{sgn}([M,N]_s)$.

$$\begin{aligned} LHS &= \int_0^\infty H_s K_s \cdot \operatorname{sgn}(H_s K_s) \cdot \operatorname{sgn}([M,N]_s) \,\mathrm{d}[M,N]_s \\ &= \int_0^\infty \tilde{H}_s K_s \,\mathrm{d}[M,N]_s \end{aligned}$$

$$RHS = \int_0^\infty \tilde{H}_s \,\mathrm{d}[M]_s \cdot \int_0^\infty K_s^2 \,\mathrm{d}[N]_s$$

$$\Leftrightarrow \left(\int_0^\infty \tilde{H}_s K_s \,\mathrm{d}[M,N]_s\right)^2 \leq \int_0^\infty \tilde{H}_s \,\mathrm{d}[M]_s \int_0^\infty K_s^2 \,\mathrm{d}[N]_s.$$

$$\Leftrightarrow \begin{aligned} H_n &= \sum_{i=1}^n U_i(t_i, t_{i+1}] \\ K_n &\equiv \sum_{i=1}^n V_i(t_i, t_{i+1}]. \end{aligned}$$

$$\Leftrightarrow \left(\sum_{i=1}^{n} U_i V_i \left([M,N]_{t_{i+1}} - [M,N]_{t_i}\right)\right)^2 \leq \sum_{i=1}^{2} U_i^2([M]_{t_{i+1}} - [M]_{t_i}) \cdot \sum_{i=1}^{n} V_i^2([N]_{t_{i+1}} - [N]_{t_i}).$$

It suffices to prove

$$\left|[M,N]_{t_{i+1}} - [M,N]_{t_i}\right| \leq \left([M]_{t_{i+1}} - [M]_{t_i}\right)^{\frac{1}{2}} \left([N]_{t_{i+1}} - [N]_{t_i}\right)^{\frac{1}{2}}.$$

like $(\sum_i a_i b_i)^2 \leq \sum_i a_i^2 \sum_i b_i^2$.

To prove this, note that $\forall\ \lambda \in \mathbb{Q}$, $[M + \lambda N]$ is increasing and so

$$[M + \lambda N]_{t_{i+1}} - [M + \lambda N]_{t_i} \geq 0 \quad \text{a.s..}$$

Also

$$[M + \lambda N]_t = [M]_t + 2\lambda[M,N]_t + \lambda^2[N]_t.$$

because $(M + \lambda N)_t^2 - [M + \lambda N]_t$ is U.I. and by the uniqueness (?)

$$M_t^2 + 2\lambda M_t N_t + \lambda^2 N_t^2 - ([M]_t + 2\lambda[M,N]_t + \lambda^2[N]_t).$$

Hence

$$\left([M]_{t_{i+1}} - [M]_{t_i}\right) + 2\lambda\left([M,N]_{t_{i+1}} - [M,N]_{t_i}\right) + \lambda^2\left([N]_{t_{i+1}} - [N]_{t_i}\right) \geq 0$$

$$\Leftrightarrow b^2 - 4ac \leq 0.$$

So

$$\left([M,N]_{t_{i+1}} - [M,N]_{t_i}\right)^2 \leq \left([M]_{t_{i+1}} - [M]_{t_i}\right)\left([N]_{t_{i+1}} - [N]_{t_i}\right).$$

$$\mathbb{E}\left[(H \cdot M)_\infty N_\infty\right] = \mathbb{E}\left[(H \cdot [M,N])_\infty\right].$$

If $H = Z(S,T]$,

$$\begin{cases} LHS = \mathbb{E}\left[Z(M_T - M_S) \cdot N_\infty\right] \\ RHS = \mathbb{E}\left[Z([M,N]_T - [M,N]_S)\right] \end{cases}$$

$$\begin{aligned} LHS &= \mathbb{E}\left[\mathbb{E}(N_\infty \mid \mathcal{F}_T) \cdot Z(M_T - M_S)\right] \\ &= \mathbb{E}\left[Z(M_T N_T - M_S N_T)\right] \\ &= \mathbb{E}[Z M_T N_T] - \mathbb{E}\left[\mathbb{E}(N_T \mid \mathcal{F}_S) \cdot Z M_S\right] \\ &= \mathbb{E}\left[Z M_T N_T\right] - \mathbb{E}[Z M_S N_S]. \end{aligned}$$

Because $M_t N_t - [M,N]_t$ is U.I.

$$\mathbb{E}[M_T N_T] = \mathbb{E}[[M,N]_T] \implies \mathbb{E}[Z(M_T N_T - [M,N]_T)] = 0.$$

And

$$\begin{aligned} LHS &= \mathbb{E}\left[Z(M_T N_T - M_S N_S)\right] \\ &= \mathbb{E}\left[Z\mathbb{E}\left[M_T N_T - M_S N_S \mid \mathcal{F}_S\right]\right] \\ &= \mathbb{E}\left[Z\mathbb{E}[[M,N]_T - [M,N]_S \mid \mathcal{F}_S]\right] \\ &= \mathbb{E}[Z([M,N]_T - [M,N]_S)]. \end{aligned}$$

Hence $\forall\ H \in b\varepsilon$

$$\mathbb{E}[(H \cdot M)_\infty \cdot N_\infty] = \mathbb{E}\left[(H \cdot [M,N])_\infty\right].$$

If $H_n \xrightarrow{L^2(M)} H$, i.e.

$$\mathbb{E}\left(\int_0^\infty (H_n(s) - H_s)^2 \,\mathrm{d}[M]_s\right) \to 0$$

$\implies (H_n \cdot M)_\infty \xrightarrow{L^2} (H \cdot M)_\infty$ by

$$\mathbb{E}[(H_n \cdot M - H \cdot M)^2_\infty] = \mathbb{E}\left[\int_0^\infty (H_n - H)\,\mathrm{d}[M]_s\right] \to 0.$$

Hence $\mathbb{E}[(H_n \cdot M)_\infty \cdot N_\infty] \to \mathbb{E}[(H \cdot M)_\infty N_\infty]$.

Next,

$$\begin{aligned} \mathbb{E}\left[|(H_n \cdot [M,N])_\infty - (H \cdot [M,N])_\infty|\right] &= \mathbb{E}\left|((H_n - H) \cdot [M,N])_\infty\right| \\ &\le \left(\mathbb{E}\left(\int_0^\infty (H_n - H)\,\mathrm{d}[M]_s\right)\right)^{\frac{1}{2}} (\mathbb{E}[N]_\infty)^{\frac{1}{2}} \to 0 \end{aligned}$$

by using

$$\left(\int_0^\infty |H_s|\,\mathrm{d}[M,N]_s\right) \le \left(\int_0^\infty H_s^2\,\mathrm{d}[M]_s\right)^{\frac{1}{2}} \left(\int_0^\infty 1\,\mathrm{d}[N]_s\right)^{\frac{1}{2}}.$$

Hence $\mathbb{E}[(H_n \cdot [M,N])_\infty] \to \mathbb{E}[(H \cdot [M,N])_\infty]$. □

Remark. proof of $[H \cdot M, N] = H \cdot [M,N] \Leftrightarrow$ It suffices to. prove $(H \cdot M)_t N_t - \int_0^\infty H_s \,\mathrm{d}[M,N]_s$ is a U.I. martingale.

$\Leftrightarrow \forall$ stopping time R

$$\mathbb{E}[(H \cdot M)_R \cdot N_R] = \mathbb{E}\left[\int_0^R H_s \,\mathrm{d}[M,N]_s\right].$$

$$\begin{aligned} LHS &= \mathbb{E}\left[(H \cdot M)^R_\infty \cdot N^R_\infty\right] \\ &= \mathbb{E}\left[(H^R \cdot M)_\infty \cdot N^R_\infty\right] \\ &= \mathbb{E}[(H^R \cdot [M, N^R])_\infty] \\ &= \mathbb{E}\left[\int_0^\infty H^R(s)\,\mathrm{d}[M,N^R]_s\right] \\ &= \mathbb{E}\left[\int_0^R H(s)\,\mathrm{d}[M,N]_s\right] = RHS. \end{aligned}$$

Definition 1.0.1 (local martingale)**.** $\mathcal{M}_{loc} = \{\text{local martingale}\}$.

$\exists$ stopping times $T_n \uparrow \infty$ s.t. M^{T_n} is a martingale.

Definition 1.0.2 (continuous local martingale)**.** $c\mathcal{M}_{loc} = \{\text{continuous local martingale}\}$.

Quadratic variation of $c\mathcal{M}_{loc}$.

If $M \in c\mathcal{M}_{loc}$, then $\exists$ $[M]$ increasing process such that $M^2 - [M] \in c\mathcal{M}_{loc}$.

Lemma 1.0.1. If $M \in c\mathcal{M}_{loc}$. Fix $n \geq 1$, define

$$T_n = \inf\{t \geq 0\colon\ |M_t| \geq n\}.$$

Then M^{T_n} is a bounded martingale.

Proof. By definition, $\exists$ stopping $(S_k) \uparrow \infty$ s.t. M^{S_k} is a martingale.

$\implies$ $M^{S_k \wedge T_n}$ is also a martingale.

So $\forall$ $s < t$,

$$\mathbb{E}(M_t^{S_k \wedge T_n} \mid \mathcal{F}_s) = M_s^{S_k \wedge T_n}.$$

Let $k \to \infty$ to get $\mathbb{E}(M_t^{T_n} \mid \mathcal{F}_s) = M_s^{T_n}$.

pf of Quadratic variation:

(Existence.) $\forall$ $n \geq 1$, we let

$$T_n = \inf\{t \geq 0\colon\ |M_t| \geq n\}.$$

Then define $\forall$ $t < T_n$,

$$[M]_t = [M^{T_n}]_t.$$

We need to check that $[M^{T_n}]_t$ is well-defined.

$$[M^{T_2}]^{T_1} = [M^{T_2}]_{t \wedge T_1}$$

Because $[M]^T = [M^T]$,

$$[M^{T_2}]^{T_1} = [M^{T_2 \wedge T_1}] = [M^{T_1}].$$

By $T_n \uparrow \infty$, we have $[M]_t$ is defined for all $t \geq 0$.

(uniqueness.) If $\exists$ A, A' increasing such that $N = M^2 - A$ and $N' = M^2 - A' \in c\mathcal{M}_{loc}$. Then $N - N' = A' - A \in c\mathcal{M}_{loc}$ is of finite variation.

Lemma 1.0.2. If $M \in c\mathcal{M}_{loc}$ and M is of finite variation, then $M_t \equiv 0$.

Proof. Let $V_M(t) =$ total variation of M on $[0, t]$. Define $T_n = \inf\{t \geq 0\colon V_M(t) > n\}$.

Then M^{T_n} is a bounded martingale and is of finite variation. Next, define

$$A_t^n \equiv \sum_{k=1}^{2^n} \left(M(t_k^n) - M(t_{k-1}^n)\right)^2$$

with $t_k^n = \frac{k}{2^n} t$.

So

$$A_t^n \leq \sup_{1 \leq k \leq 2^n} \left| M(t_k^n) - M(t_{k-1}^n) \right| \cdot V_M(t) \xrightarrow{a.s.} 0$$

by continuity of M.

Thus $A_t^n \to 0$, $\forall\ t \geq 0$.

We will prove $A_t^n \to A_t$ such that $M_t^2 - A_t$ is martingale.

$\implies A_t = 0$ and $\mathbb{E} M_t^2 = \mathbb{E} M_0^2 = 0$.

$\implies M_t = 0$ a.s. $\forall\ t \geq 0$.

By a.s. continuity, $M_t \equiv 0$, a.s. □

□

1.0.1 Itô Integral of local martingale

Theorem 1.0.3. Let $M \in c\mathcal{M}_{loc}$ with $M_0 = 0$. Let $H \in \mathcal{P}$ such that W.P.1

$$\int_0^t H_s^2 \, \mathrm{d}[M]_s < \infty$$

for all $t > 0$. Then $H \cdot M$ is well-defined such that $H \cdot M \in c\mathcal{M}_{0,loc}$.

Proof. Let $U_n = \inf\{t\colon\ |M_t| > n\}$, $V_n = \inf\{t\colon\ \int_0^t H_s^2 \, \mathrm{d}[M]_s > n\}$.

(Recall $L^2(M) \implies \mathbb{E} \int_0^\infty H_s^2 \, \mathrm{d}[M]_s < \infty$).

$\implies T_n = U_n \wedge V_n$. In this way, $M^{T_n} \in c\mathcal{M}_0^2$ and $H^{T_n} \in L^2(M^{T_n})$.

So $H^{T_n} \cdot M^{T_n}$ is defined by previous L^2 theory.

$\implies (H \cdot M)^{T_n} = H^{T_n} \cdot M^{T_n}$ $(t < T_n)$ is a L^2-bounded martingale.

$\implies H \cdot M \in c\mathcal{M}_{0,loc}$. □

Definition 1.0.3 (semimartingale)**.** We say X is a semimartingale if

$$X_t = X_0 + M_t + A_t.$$

M_t is local mart, A_t is finite variation.

Definition 1.0.4. Let X be a continuous semimartingale. Then $\exists$ unique

$$\begin{cases} M \in c\mathcal{M}_{0,loc} \\ A \in cFV_0 \end{cases}$$

such that $X_t = X_0 + M_t + A_t$, $\forall\ t \geq 0$.

Proof. If $\exists\ M'$ and A' such that

$$X = X_0 + M' + A' = X_0 + M + A.$$

Then $M - M' = A' - A \in c\mathcal{M}_{0,loc} \cap cFV = \{0\}$. $\square$

We write $M = X^{cm}$ and

$$[X, X] = [M, M] = [X^{cm}, X^{cm}].$$

Similarly, $[X, Y] = [X^{cm}, Y^{cm}]$.

For any $H \in lb\mathcal{P}$ define

$$H \cdot X = H \cdot M + H \cdot A.$$

Obviously $H \cdot M \in c\mathcal{M}_{0,loc}$ and $H \cdot A \in FV$.

And

$$\begin{aligned}
[H \cdot X, K \cdot Y] &= [H \cdot X^{cm}, K \cdot Y^{cm}] \\
&= H \cdot [X^{cm}, K \cdot Y^{cm}] \\
&= H \cdot K[X^{cm}, Y^{cm}] \\
&= (HK) \cdot [X^{cm}, Y^{cm}] = (HK) \cdot [X, Y].
\end{aligned}$$

1.0.2 Itô's Formula

Theorem 1.0.4 (Itô's Formula)**.** Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be C^2. Let

$$X_t \equiv (X_t^1, \cdots, X_t^n)$$

be a continuous semimartingale.

Then

$$f(x_t) - f(x_0) = \int_0^t \nabla f(X_s)\,\mathrm{d}X_s + \frac{1}{2}\int_0^t \sum_{i,j} D_{ij} f(X_s)\,\mathrm{d}[X^i, X^j]_s$$

where $\nabla f(X_s) = (D_1 f(X_s), \cdots, D_n f(X_s))$. $D_i f(x) = \frac{\partial}{\partial x_i} f(x)$, $D_{ij} = D_i D_j = D_j D_i$.

Theorem 1.0.5 (Integration by Parts)**.** Let X, Y be continuous semimartingales. Then

$$X_t Y_t - X_0 Y_0 = \int_0^t X_s\,\mathrm{d}Y_s + \int_0^t Y_s\,\mathrm{d}X_s + [X, Y]_t.$$

Proof. Assume IBP holds.

Let

$$\mathcal{A} = \left\{ f \in C^2(\mathbb{R}^n)\colon\ \mathrm{d}f(X_t) = \sum_i D_i f(X_t)\,\mathrm{d}X_t + \frac{1}{2}\sum_{i,j} D_{ij} f(X_t)\,\mathrm{d}[X^i, X^j]_t \right\}.$$

(i). $f, g \in \mathcal{A} \implies f + g \in \mathcal{A}$. (ii). $f, g \in \mathcal{A} \implies fg \in \mathcal{A}$.

$$
\begin{aligned}
\mathrm{d}(fg(X_t)) =& \mathrm{d}(f(X_t)\cdot g(X_t)) \\
=& f(X_t)\cdot \mathrm{d}g(X_t) + g(X_t)\,\mathrm{d}f(X_t) + \mathrm{d}[f(X_t), g(X_t)]_t \\
=& f(X_t)\left(\sum_i D_i g(X_t)\,\mathrm{d}X_t + \frac{1}{2}\sum_{i,j} D_{ij} g(X_t)\,\mathrm{d}[X^i, X^j]_t\right) \\
&+ g(X_t)\left(\sum_i D_i f(X_t)\,\mathrm{d}X_t + \frac{1}{2}\sum_{i,j} D_{ij} f(X_t)\,\mathrm{d}[X^i, X^j]_t\right) \\
&+ \mathrm{d}\left[\int_0^t \sum_i D_i f(X_t)\,\mathrm{d}X_t, \int_0^t \sum_j D_j g(X_t)\,\mathrm{d}X_t^j\right] \\
=& \mathrm{d}X_t\left(\sum_i f D_i g(X_t) + \sum_i g D_i f(X_t)\right) \\
&+ \frac{1}{2}\sum_{i,j}\left(g D_{ij} f(X_t) + f D_{ij} g(X_t) + 2 D_i f D_j g(X_t)\right)\mathrm{d}[X^i, X^j]_t \\
=& \sum_i D_i(fg)(X_t)\,\mathrm{d}X_t + \frac{1}{2}\sum_{i,j} D_{ij}(fg)(X_t)\,\mathrm{d}[X^i, X^j]_t \\
\implies& \quad fg \in \mathcal{A}.
\end{aligned}
$$

(iii). $f(x) = x \in \mathcal{A} \implies \forall$ polynomial $\in \mathcal{A}$. □

Lemma 1.0.3. $\forall\ f \in C^2(\mathbb{R}^d)$, $\exists$ polynomials $\{P_n(x)\}_{n\geq 1}$ such that

$$
\begin{cases}
P_n(x) \to f(x) \\
D_i P_n(x) \to D_i f(x) \\
D_{ij} P_n(x) \to D_{ij} f(x)
\end{cases}
$$

uniformly on compacts.

Define $U_N = \inf\{t \geq 0\colon\ |X_t| + [M]_t > N\}$ for each $N > 0$.
Then $U_N \uparrow \infty$. So it suffices to prove Itô formula holds for $\forall\ t \leq U_N$.
Note that

$$
P_n(X_t) = P_n(X_0) + \int_0^t \sum_i D_i P_n(X_s)\,\mathrm{d}X_s + \frac{1}{2}\int_0^t \sum_{i,j} D_{ij} P_n(X_s)\,\mathrm{d}[X^i, X^j]_s
$$

for $0 \leq t \leq U_N$.

By letting $n \to \infty$, we will prove the limits are

$$
f(X_t) = f(X_0) + \int_0^t \sum_i D_i f(X_s)\,\mathrm{d}X_s + \frac{1}{2}\int_0^t \sum_{ij} D_{ij} f(X_s)\,\mathrm{d}[X^i, X^j]_s.
$$

$$
\begin{aligned}
I_{1,n} &= \int_0^t \sum_i D_i P_n(X_s)\,\mathrm{d}X_s \\
&= \int_0^t \sum_i D_i P_n(X_s)\,\mathrm{d}M_s + \int_0^t \sum_i D_i P_n(X_s)\,\mathrm{d}A_s.
\end{aligned}
$$

Notice that

$$
\left| \int_0^t \sum_i D_i P_n(X_s)\,\mathrm{d}A_s - \int_0^t \sum_i D_i f(X_s)\,\mathrm{d}A_s \right| \le d \cdot \sup_i \underbrace{\|D_i P_n - D_i f\|}_{\text{max on compacts}} \cdot A_t \xrightarrow{n\to\infty} 0 \text{ a.s.}
$$

Next,

$$
\begin{aligned}
&\mathbb{E} \sup_{s\le t} \left| \int_0^t \sum_i \left(D_i P_n(X_s) - D_i f(X_s)\right) \mathrm{d}M_s \right|^2 \\
\le &\mathbb{E} \int_0^t \left(\sum_i D_i P_n(X_s) - D_i f(X_s) \right)^2 \mathrm{d}[M]_s \\
\le &d \cdot \sup_{1\le i\le d} \|D_i P_n - D_i f\|^2 \cdot \underbrace{\mathbb{E}[M]_{t\wedge U_N}}_{N} \xrightarrow{n\to\infty} 0.
\end{aligned}
$$

By taking subsequence, $\int_0^t \sum_i D_i P_{n_k}(X_s)\,\mathrm{d}M_s \xleftarrow{\text{a.s.}} \int_0^t \sum_i D_i f(X_s)\,\mathrm{d}M_s$.

$$
I_{2,n} = \frac{1}{2} \int_0^t \sum_{i,j} D_{ij} P_n(X_s)\,\mathrm{d}[X^i, X^j]_s.
$$

Notice that

$$
\left| I_{2,n} - \frac{1}{2} \int_0^t \sum_{i,j} D_{ij} f(X_s)\,\mathrm{d}[X^i, X^j]_s \right| \le C \sum_{i,j} \|D_{ij} P_n - D_{ij} f\| \cdot \underbrace{\sum_{i,j} [X^i, X^j]_t}_{\le C_N} \xrightarrow{n\to\infty} \to 0.
$$

Lemma 1.0.4. Let M be a bounded continuous martingale with $M_0 = 0$. Let V be continuous FV_0

$$
\begin{aligned}
M_t^2 &= \int_0^t 2M_s\,\mathrm{d}M_s + [M]_t, \\
M_t V_t &= \int_0^t M_s\,\mathrm{d}V_s + \int_0^t V_s\,\mathrm{d}M_s.
\end{aligned}
$$

Proof.

$$\begin{aligned}
M_t^2 &= \left[\sum_{k=1}^{2^n} M(t_k^n) - M(t_{k-1}^n)\right]^2 \\
&= \sum_{k=1}^{2^n} \left(M(t_k^n) - M(t_{k-1}^n)\right)^2 + 2\sum_{k=1}^{2^n} \left[M(t_k^n)M(t_{k-1}^n) - M(t_{k-1}^n)^2\right] \\
&= M(t_{2^n}^n)^2 - M(t_0^n)^2 = M(t)^2
\end{aligned}$$

where $t_k^n = \frac{k}{2^n}t$.

Note that $\sum_{k=1}^{2^n} \left(M(t_k^n) - M(t_{k-1}^n)\right)^2 \to [M]_t$ and

$$2\sum_{k=1}^{2^n} \left[M(t_k^n)M(t_{k-1}^n) - M(t_{k-1}^n)^2\right] = 2\sum_{k=1}^{2^n} M(t_{k-1}^n)(M(t_k^n) - M(t_{k-1}^n)) \to 2\int_0^t M_s \,\mathrm{d}M_s.$$

$$\begin{aligned}
M_tV_t &= M_{t_{2^n}^n}V_{t_{2^n}^n} - M_{t_0^n}V_{t_0^n} \\
&= \sum_{k=1}^{2^n} \left[M(t_k^n)V(t_k^n) - M(t_{k-1}^n)V(t_{k-1}^n)\right] \\
&= \sum_{k=1}^{2^n} M(t_k^n)(V(t_k^n) - V(t_{k-1}^n)) + \sum_{k=1}^{n} V(t_{k-1}^n)\left(M(t_k^n) - M(t_{k-1}^n)\right) \\
&\to \int_0^t M_s \,\mathrm{d}V_s + \int_0^t V_s \,\mathrm{d}M_s.
\end{aligned}$$

□

Theorem 1.0.6 (Lévy Thm)**.** Let $\{B_t^i\}_{t\geq 0} \in c\mathcal{M}_{0,loc}$ for $i = 1, 2, \cdots, n$. If $\forall\ i, j$, we have

$$[B^i, B^j]_t = \begin{cases} t, & i = j, \\ 0, & i \neq j. \end{cases}$$

Then $B_t = (B_t^1, \cdots, B_t^n)$ is a Brownian motion on $\mathbb{R}^n$ with $\mathcal{N}(0, I_n)$.

Proof. $\forall\ s < t$, we will prove

$$\mathbb{E}\left[e^{i\theta(B_t - B_s)}\middle|\mathcal{F}_s\right] = e^{-\frac{1}{2}(t-s)|\theta|^2}$$

for all $\theta \in \mathbb{R}^n$.

Define $f(x, t) = e^{i\theta x + \frac{1}{2}|\theta|^2 t}$, $\mathbb{R}^{n+1} \to \mathbb{C}$.

Apply Itô's formula to $f(B_t, t) \equiv f(X_t)$.

$X_t = (B_t^1, B_t^2, \cdots, B_t^n, t)$.

$$
\begin{aligned}
f(X_t) =& f(x_0) + \int_0^t \sum_{i=1}^n (D_i f(X_s \, \mathrm{d}B_s^i)) + \int_0^1 D_{n+1} f(X_s) ds \\
&+ \frac{1}{2} \int_0^t \sum_{1 \le i,j \le n} D_{ij} f(X_s) \, \mathrm{d}[B^i, B^j]_s \\
&+ \frac{1}{2} \int_0^t \sum_{1 \le k \le n} D_{k,n+1} f(X_s) \, \mathrm{d}[B^k, s]_s \\
\implies f(B_t, t) =& f(B_0, 0) + \int_0^t \sum_i D_i f(B_s, s) \, \mathrm{d}B_s^i \\
&+ \int_0^t \left(e^{i\theta B_s + \frac{1}{2}|\theta|^2 s} \cdot \frac{1}{2} |\theta|^2 \right) \mathrm{d}s \\
&+ \int_0^t \frac{1}{2} \left(\sum_{j=1}^n e^{i\theta B_s + \frac{1}{2}|\theta|^2 s} (i\theta_j)^2 \right) \mathrm{d}s.
\end{aligned}
$$

We conclude that $f(B_t, t) = f(B_0, 0) + c\mathcal{M}_{0,loc}$

$$
\implies e^{\underbrace{i\theta B_t + \frac{1}{2}|\theta|^2 t}_{M_t^\theta \in c\mathcal{M}_{0,loc}}} = 1 + c\mathcal{M}_{0,loc}
$$

Since $\left|M_t^\theta\right| = e^{\frac{1}{2}|\theta|^2 t} < \infty$ and

$$
\mathbb{E}(M_t^{T_n} \mid \mathcal{F}_s) = M_s^{T_n}
$$

where T_n reduces M.

We may take $n \to \infty$ to get $\mathbb{E}(M_t \mid \mathcal{F}_s) = M_s$.

$$
\implies \mathbb{E}(e^{i\theta B_t + \frac{1}{2}|\theta|^2 t} \mid \mathcal{F}_s) = e^{i\theta B_s + \frac{1}{2}|\theta|^2 s}.
$$

$$
\implies \mathbb{E}\left(e^{i\theta(B_t - B_s)} \mid \mathcal{F}_s\right) = e^{\frac{1}{2}|\theta|^2 (t-s)}.
$$

$$
\implies B_t - B_s \sim \mathcal{N}(0, t-s)
$$

Q independent of $\mathcal{F}_s$. □

Theorem 1.0.7 (Dubins and Schwarz)**.** let $M \in c\mathcal{M}_{0,loc}$ such that $[M]_t \uparrow \infty$ as $t \to \infty$. Then $M_t = B([M]_t)$ where B is some Brownian motion.

Let $\tau_t = \inf\{u \ge 0 \colon [M]_u > t\}$. Set $\mathcal{G}_t \equiv \mathcal{F}(\tau_t)$. Then $B_t \equiv M(\tau_t)$ is a B.M. with respect to $\mathcal{G}_t$.

Proof. (i) $M(\tau_t) \in \mathcal{F}_{\tau_t} \implies B_t \in \mathcal{G}_t$.

(ii) $t \mapsto B_t$ is continuous.

$$\begin{cases} \tau_{a-} = b, \\ \tau_{a+} = c. \end{cases}$$

We only need to prove $M(b) = M(c)$.
Assume M is bounded $\forall\ q \in \mathbb{Q}_+$, define

$$S_q \equiv \inf\{t > q \colon [M]_t > [M]_q\}.$$

We will show $M_t \equiv M_q$ for all $t \in [q, S_q)$
Note $S_q \geq q$, by OST,

$$\mathbb{E}\left(M_{S_q}^2 - [M]_{S_q} \mid \mathcal{F}_q\right) = M_q^2 - [M]_q.$$

Because $[M]_{S_q} = [M]_q$,

$$\implies \mathbb{E}[M_{S_q}^2 - M_q^2 \mid \mathcal{F}_q] = 0.$$

$$\implies \mathbb{E}((M_{S_q} - M_q)^2 \mid \mathcal{F}_q) = 0.$$

$$\implies \mathbb{E}(M_{S_q} - M_q)^2 = 0.$$

$$\implies M_{S_q} = M_q.$$

(iii) B_t and $B_t^2 - t \in c\mathcal{M}_{0,loc} \implies [B]_t = t \implies$ B.M.
Recall $B_t = M(\tau_t)$. Define

$$T_n = \inf\{t \geq 0 \colon |M_t| > n\},$$
$$U_n = [M]_{T_n}.$$

We will check $\tau_{t\wedge U_n} = T_n \wedge \tau_t$.

$$t \wedge U_n \leq t \implies \tau_{t\wedge U_n} \leq \tau_t.$$

$$\tau_{t\wedge U_n} = \inf\{u \colon [M]_u > t \wedge U_n = t \wedge [M]_{T_n}\}.$$

Claim: $\tau_t \wedge T_n = \tau_{t\wedge U_n}$
case 1. $t \leq U_n$ we have

$$\tau_{t\wedge U_n} = \inf\{u \colon [M]_u > t\} = \tau_t,$$

and

$$t \leq U_n = [M]_{T_n} \implies T_n \geq \tau_t \implies \tau_t \wedge T_n = \tau_t = \tau_{t\wedge U_n}.$$

case 2. $t > U_n$ we have

$$\tau_{t\wedge U_n} = \tau_{U_n} = \inf\{u \colon [M]_u > U_n\}$$

and

$$t > U_n > [M]_{T_n} \implies T_n < \tau_t \implies \tau_t \wedge T_n = T_n.$$

It suffices to prove $T_n = \tau_{[M]_{T_n}} = \tau_{U_n}$.

One hand, $\tau_{U_n} \geq T_n$ can be proved by def of τ_{U_n}.

On the other hand, $T_n = \inf\{t\colon |M_t| > n\}$. Take $\forall\ \varepsilon > 0$, $u_\varepsilon \in (T_n, T_n + \varepsilon)$ s.t. $|M_{u_\varepsilon}| > n$.

Hence $M_{u_\varepsilon} \neq M_{T_n}$. Suppose $[M]_{u_\varepsilon} = [M]_{T_n}$, then M is constant by the property of continuous local martingale. Contradiction. So $[M]_{u_\varepsilon} > [M]_{T_n} = U_n \implies \tau_{U_n} \leq u_\varepsilon$. Let $\varepsilon \downarrow 0$ to get $\tau_{U_n} \leq T_n$.

Then $B_{t\wedge U_n} = M(\tau_{t\wedge U_n}) = M(T_n \wedge \tau_t)$, $B_t^{U_n} = M_{\tau_t}^{T_n}$.

By OST

$$\mathbb{E}\left(B_t^{U_n}\middle|\mathcal{F}_{\tau_s}\right) = \mathbb{E}\left(M_{\tau_t}^{T_n}\middle|\mathcal{F}_{\tau_s}\right) = M_{\tau_s}^{T_n} = B_s^{U_n}.$$

Then B^{U_n} is a martingale w.r.t. $\mathcal{G}_t = \mathcal{F}_{\tau_t}$ + (2) U_n is $\mathcal{G}_t$-stopping time $\implies B \in c\mathcal{M}_{0,loc}$

pf of (2)

$$\{[M]_{T_n} \leq t\} = \{U_n \leq t\} \in \mathcal{F}_{\tau_t}.$$

$$\{T_n \leq \tau_t\} \in \mathcal{F}_{\tau_t}.$$

$\left(M^2 - [M]\right)^{T_n}$ is martingale. Then

$$\mathbb{E}\left[(M_{\tau_t}^{T_n})^2 - [M]_{\tau_t}^{T_n}\middle|\mathcal{F}_{\tau_s}\right] = \left(M_{\tau_s}^{T_n}\right)^2 - [M]_{\tau_s}^{T_n}$$

$$\implies \mathbb{E}\left[(B_t^{U_n})^2 - t\wedge U_n\middle|\mathcal{F}_s\right] = (B_s^{U_n})^2 - s\wedge U_n.$$

$\implies (B_t^2 - t)^{U_n}$ is a martingale $\implies B_t^2 - t \in \mathcal{M}_{loc} \implies [B]_t = t.$ □

If $[M]_\infty < \infty$, we get $\tau_t = \infty$, $\forall\ t \geq [M]_\infty$.

$$B_t = M(\infty), \quad \forall\ t \geq [M]_\infty.$$

So this def does not give a B.M.

Define

$$B_t = \begin{cases} M(\tau_t), & t < [M]_\infty, \\ M(\infty) + W(t) - W([M]_\infty), & t \geq [M]_\infty, \end{cases}$$

where W is a standard B.M. independent of M.

Or

$$B_t \begin{cases} M(\tau_t), & t < [M]_\infty \\ M(\infty) + W(t - [M]_\infty), & t \geq [M]_\infty. \end{cases}$$

Then $B([M]_t) = M_t$.

Lemma 1.0.5. On $\{[M]_\infty < \infty\}$, the limit $M_\infty \equiv \lim_{t\uparrow\infty} M_t$ exists a.s.

Proof. Define $S_K = \inf\{t \geq 0\colon [M]_t > K\}$. Then on $\{[M]_\infty < \infty\}$, $S_K = \infty$, $\forall\ K > [M]_\infty$

M^{S_K} is a L^2-bounded martingale.

$\implies \lim_{t\uparrow\infty} M_t^{S_K}$ exists a.s.

Let $K \uparrow \infty$ to get $\lim_{t\uparrow\infty} M_t$ exists a.s.

By an enlargement of the prob. space $(\Omega.\mathcal{F}, \mathsf{P})$

$$\begin{aligned} \Longrightarrow \tilde{\Omega} &= \Omega \times \Omega^* \\ \tilde{\mathcal{F}}_t &= \mathcal{F}_t \times \mathcal{F}_t^* \\ \tilde{\mathsf{P}} &= \mathsf{P} \times \mathsf{P}^* \end{aligned}$$

where on $(\Omega^*, \mathcal{F}^*, \mathcal{F}_t^*, \mathsf{P}^*)$ we have W^* is B.M. independent of M. □

Theorem 1.0.8 (Dubins-Schwarz (general version))**.** Let $M \in c\mathcal{M}_{0,loc}$. Then $\exists$ B.M. B s.t. $M_t = B([M]_t)$ for all $t \geq 0$.

Theorem 1.0.9 (Doléans' Thm)**.** Let X be a semimartingale with $X_0 = 0$. Suppose $Z_0 \in \mathcal{F}_0$. Then $\exists$ a unique continuous semimartingale Z s.t.

$$Z_t = Z_0 + \int_0^t Z_s \, \mathrm{d}X_s.$$

The solution $Z_t = Z_0 e^{X_t - \frac{1}{2}[X]_t}$.

Notation: $\mathcal{E}(X_t) = e^{X_t - \frac{1}{2}[X]_t}$ is called the exponential semimartingale of X.

Proof. $\mathcal{E}(X_t) = e^{X_t - \frac{1}{2}[X]_t} = f(X_t, [X]_t)$ with $f(x, y) = e^{x - \frac{1}{2}y}$.

By apply Itô's formula,

$$\mathrm{d}f(X_t, [X]_t) = \partial_x f(X_t, [X]_t) \, \mathrm{d}X_t + \partial_y f(X_t, [X]_t) \, \mathrm{d}[X]_t + \frac{1}{2} \left(\partial_{xx} f(X_t, [X]_t) \, \mathrm{d}[X]_t \right).$$

Because

$$\begin{cases} \mathrm{d}X_t \, \mathrm{d}X_t = \mathrm{d}[X]_t \\ \mathrm{d}X_t \, \mathrm{d}[X]_t = 0 \\ \mathrm{d}[X]_t \, \mathrm{d}[X]_t = 0. \end{cases}$$

Hence

$$\mathrm{d}\mathcal{E}(X_t) = \mathcal{E}(X_t) \, \mathrm{d}X_t.$$

So $\mathcal{E}(X)$ is a solution to

$$\mathrm{d}Z_t = Z_t \, \mathrm{d}X_t.$$

(Uniqueness) Assume Z is another solution. We will prove

$$Z_t = Z_0 \mathcal{E}(X_t).$$

We now have $\mathrm{d}Z_t = Z_t \, \mathrm{d}X_t$.

Define

$$Y_t = e^{-X_t + \frac{1}{2}[X]_t} = (\mathcal{E}(X)_t)^{-1}.$$

It suffices to prove $\mathrm{d}(Z_tY_t) = 0$ which implies

$$Z_tY_t = Z_0Y_0 = Z_0, \quad \forall\ t.$$

We have

$$\mathrm{d}(Z_tY_t) = Z_t\,\mathrm{d}Y_t + Y_t\,\mathrm{d}Z_t + \mathrm{d}[Y,Z]_t.$$

Notice that by Itô's formula,

$$\mathrm{d}Y_t = (-1)Y_t\,\mathrm{d}X_t + \frac{1}{2}Y_t\,\mathrm{d}[X]_t + \frac{1}{2}(-1)^2Y_t\,\mathrm{d}[X]_t = -Y_t\,\mathrm{d}X_t + Y_t\,\mathrm{d}[X]_t.$$

$$\implies \mathrm{d}(Z_tY_t) = Z_t(-Y_t\,\mathrm{d}X_t + Y_t\,\mathrm{d}[X]_t) + Y_t(Z_t\,\mathrm{d}X_t) + Y_t(Z_t\,\mathrm{d}X_t) + (-Y_tZ_t)\,\mathrm{d}[X]_t = 0.$$

□

Let $M \in c\mathcal{M}_{0,loc}$. Then $\mathcal{E}(\theta M)_t = e^{\theta M_t - \frac{1}{2}\theta^2[M]_t}$ is in $c\mathcal{M}_{loc}$.
By Fatou's lemma, we get

$$\begin{aligned}
\mathbb{E}(\mathcal{E}(\theta M)_t \mid \mathcal{F}_s) &= \mathbb{E}(\lim_{n\to\infty} \mathcal{E}(\theta M)_{t\wedge T_n} \mid \mathcal{F}_s) \\
&\le \liminf_{n\to\infty} \mathbb{E}(\mathcal{E}(\theta M)_t^{T_n} \mid \mathcal{F}_s) \\
&= \liminf_{n\to\infty} \mathcal{E}(\theta M)_s^{T_n} \\
&= \mathcal{E}(\theta M)_s.
\end{aligned}$$

Hence $\mathcal{E}(X_t)$ is a supermartingale and $\mathbb{E}(\mathcal{E}(M)_t) \le 1$.

Theorem 1.0.10. Suppose $\forall\ t > 0$, $\exists\ K_t > 0$ s.t.

$$[M]_t < K_t \quad \text{a.s.}$$

If for some $t > 0$, $\exists\ K_t > 0$ s.t.

$$[M]_t < K_t \quad \text{a.s.}$$

Then for this $t > 0$,

$$\mathsf{P}\left(\max_{s\le t} M_s > y\right) \le e^{-\frac{y^2}{2K_t}}, \quad \forall\ y > 0.$$

and $\mathcal{E}(\theta M)$ is a true martingale.

Proof. By using $\mathcal{E}(\theta M)$ is a supermartingale, by Doob's inequality,

$$\begin{aligned}
\mathsf{P}\left(\max_{s\le t} M_s > y\right) &= \mathsf{P}\left(\max_{s\le t} \mathcal{E}(\theta M)_s > e^{\theta y - \frac{1}{2}\theta^2 K_t}\right) \\
&\le \frac{\mathbb{E}(\mathcal{E}(\theta M)_t)}{e^{\theta y - \frac{1}{2}\theta^2 K_t}} \\
&\le e^{-\theta y + \frac{1}{2}\theta^2 K_t}.
\end{aligned}$$

Pick $\theta = \frac{y}{K_t}$ to get $e^{-\frac{y^2}{2K_t}}$.

Next, define $Z_t = \max_{s\le t} M_s$. Then $\forall\ \theta > 0$,

$$\mathbb{E}e^{\theta Z_t} = \int_{\mathbb{R}} e^{\theta y}\mathsf{P}(Z_t \in \mathrm{d}y).$$

Notice that $\forall\ z \in \mathbb{R}$,

$$e^{\theta Z} = 1 + \int_0^Z \theta e^{\theta y}\,\mathrm{d}y.$$

$$\begin{aligned}
\implies \mathbb{E}(e^{\theta Z_t}) &= 1 + \mathbb{E}\left(\int_0^{Z_t} \theta e^{\theta y}\,\mathrm{d}y\right) \\
&= 1 + \mathbb{E}\left(\int_0^{\infty} \mathsf{1}_{y<Z_t}\theta e^{\theta y}\,\mathrm{d}y\right) \\
&= 1 + \int_0^{\infty} \theta e^{\theta y}\,\mathrm{d}y\mathsf{P}(Z_t > y) \\
&\le 1 + \int_0^{\infty} \theta e^{\theta y} e^{-\frac{y^2}{2K_t}}\,\mathrm{d}y \\
&< \infty.
\end{aligned}$$

By $\mathcal{E}(\theta M)_t$ is a $\mathcal{M}_{loc}$, we get $\exists\ T_n \uparrow \infty$ s.t. $\mathcal{E}(\theta M)_t^{T_n}$ is a martingale.

$$\implies \forall\ s < t, \quad \mathbb{E}[\mathcal{E}(\theta M)_t^{T_n} \mid \mathcal{F}_s] = \mathcal{E}(\theta M)_s^{T_n}.$$

By DCT for $\mathcal{E}(\theta M)_t^{T_n} \le Z_t$ and $\mathbb{E}Z_t < \infty$, we get

$$\lim_{n\to\infty} \mathbb{E}(\mathcal{E}(\theta M)_t^{T_n} \mid \mathcal{F}_s) = \mathbb{E}(\mathcal{E}(\theta M)_t \mid \mathcal{F}_s).$$

□

Corollary 1.0.1. Let B be a B.M. in $\mathbb{R}^n$ and let H be a bounded previsible process. Then

$$e^{\int H\,\mathrm{d}B - \frac{1}{2}\int |H_s|^2\,\mathrm{d}s}$$

is a true martingale.

Remark. $[M]_t = \int_0^t |H_s|^2\,\mathrm{d}s < K_t$.

Corollary 1.0.2. Let $M \in c\mathcal{M}_{0,loc}$. Then $\forall\ t > 0,\ y > 0,\ K > 0$,

$$\mathsf{P}\left(\max_{s\le t} M_s \ge y, [M]_t \le K\right) \le e^{-\frac{y^2}{2K}}.$$

$$\implies \mathsf{P}\left(\max_{s\le t} |M_s| \ge y, [M]_t \le K\right) \le 2e^{-\frac{y^2}{2K}}.$$

Proof. Let $T = \inf\{u \ge 0 \colon [M]_u \ge K\}$. Then

$$\mathsf{P}\left(\max_{s\le t} M_s \ge y, [M]_t \le K\right) \le \mathsf{P}\left(\max_{s\le t} M_{s\wedge T} \ge y\right) \le e^{-\frac{y^2}{2K}}$$

since $[M^T]_t \le K$. □

Theorem 1.0.11 (Novikov's Thm)**.** If $\forall\ t > 0$, $\mathbb{E}(e^{\frac{1}{2}[M]_t}) < \infty$ then $\mathcal{E}(X)_t$ is a true martingale.

Proof. See P198 of [KS]. □

1.0.3 Burkholder-Davis-Gundy inequalities

Definition 1.0.5. We say a function $F : \mathbb{R}^+ \to \mathbb{R}^+$ is <u>moderate</u> if continuous and increasing, $F(x) = 0 \Leftrightarrow x = 0$, and $\forall\ \alpha > 1$,

$$\sup_{x>0} \frac{F(\alpha x)}{F(x)} < \infty.$$

Example . $F(x) = x^p$, $\forall\ p > 0$. $F(x) = x^p\,(\log x \vee 1)$, $\forall\ p \ge 0$.

Theorem 1.0.12 (BDG inequality)**.** Let F be moderate. Then $\exists$ universal constants $c_F \le C_F < \infty$ s.t. $\forall\ M \in c\mathcal{M}_{0,loc}$,

$$c_F\mathbb{E}\left(F([M]_\infty^{\frac{1}{2}})\right) \le \mathbb{E}F(M_\infty^*) \le C_F\mathbb{E}\left(F([M]_\infty^{\frac{1}{2}})\right).$$

$M_t^* = \sup_{s\le t} |M_s|$. Fix $t > 0$. Let $\tilde{M}_s^t = M_{s\wedge t}$. Then

$$c_F\mathbb{E}\left(F([M]_t^{\frac{1}{2}})\right) \le \mathbb{E}\left(F(M_t^*)\right) \le C_F\mathbb{E}F([M]_t^{\frac{1}{2}}).$$

Example . If $F(x) = x^p$, then

$$c_p\mathbb{E}[M]_t^{\frac{p}{2}} \le \mathbb{E}\,|M_t^*|^p \le C_p\mathbb{E}[M]_t^{\frac{p}{2}}.$$

Lemma 1.0.6 ("good" λ inequality)**.** Let X and Y be nonegative R.V.s. Suppose $\exists\ \beta > 1$ s.t. $\forall\ \lambda > 0,\ \delta > 0$,

$$\mathsf{P}(X > \beta\lambda, Y \le \delta\lambda) \le \psi(\delta)\mathsf{P}(X > \lambda)$$

where $\psi(\delta) \downarrow 0$ as $\delta \downarrow 0$.

Then for each moderate F, $\exists\ C = C(\beta, \psi, F) > 0$ s.t.

$$\mathbb{E}[F(X)] \leq c\mathbb{E}[F(Y)].$$

Proof for Lemma.

- Assume good λ inequality. Then it holds by replacing X by $X \wedge a$ for any $a > 0$.
  If $a > \beta\lambda > \lambda$, then

$$X \wedge a > \beta\lambda \Leftrightarrow X > \beta\lambda$$
$$X \wedge a > \lambda \Leftrightarrow X > \lambda.$$

  Hence we may assume $\mathbb{E}F(X) < \infty$.
  [$\mathbb{E}F(X \wedge a) \leq \mathbb{E}F(Y)$ for all $a > 0$. Then let $a \uparrow \infty$.]
  Notice that $\sup_{x>0} \frac{F(x)}{F(\frac{x}{\beta})}$.
  Let $\gamma > 0$ s.t. $F(x) \leq \frac{1}{\gamma} F(\frac{x}{\beta})$ for all $x > 0$.
  Now

$$\int F(\mathrm{d}\lambda)\psi(\delta)\mathsf{P}(X > \lambda) \geq \int F(\mathrm{d}\lambda)\mathsf{P}(X > \beta\lambda, Y \leq \delta\lambda).$$

$$LHS = \psi(\delta)\mathbb{E}\left(\int_0^X F(\mathrm{d}\lambda)\right) = \psi(\delta)\mathbb{E}(F(X)).$$

$$\begin{aligned} RHS &= \int F(\mathrm{d}\lambda)\mathbb{E}(1_{\frac{Y}{\delta} \leq \lambda < \frac{X}{\beta}}) \\ &= \mathbb{E}\left(\int_{\frac{Y}{\delta}}^{\frac{X}{\beta}} F(\mathrm{d}\lambda)\right) \\ &= \mathbb{E}\left(F\left(\frac{X}{\beta}\right) - F\left(\frac{Y}{\delta}\right)\right) \\ &\geq \mathbb{E}\left(\gamma F(X)\right) - \mathbb{E}\left(F\left(\frac{Y}{\delta}\right)\right). \end{aligned}$$

  Let $\delta > 0$ be small sucht that

$$\psi(\delta) < \frac{\gamma}{2} \implies \gamma - \psi(\delta) \geq \frac{\gamma}{2}.$$

  Hence

$$\mathbb{E}\left(F(X)\right)\left(\psi(\delta) - \gamma\right) \geq -\mathbb{E}\left(F\left(\frac{Y}{\delta}\right)\right).$$

$$\implies \mathbb{E}\left(F\left(\frac{Y}{\delta}\right)\right) \geq \left(\gamma - \psi(\delta)\right)\mathbb{E}F(X) \geq \frac{\lambda}{2}\mathbb{E}F(X).$$

  Using $\sup\limits_{y>0} \dfrac{F(y/\delta)}{F(y)} < \infty$, $\implies \exists\ \mu > 0$ s.t. $F(y/\delta) \leq \mu F(y)$ for all $y > 0$

$$\implies \mathbb{E}(F(Y/\delta)) \leq \mu\mathbb{E}F(Y)$$

$$\implies \frac{\gamma}{2}\mathbb{E}F(X) \le \mu\mathbb{E}F(Y).$$

□

Proof. of BDF inequality.

$$\mathsf{P}\left(M_\infty^* > \beta\lambda, [M]_\infty^{\frac{1}{2}} \le \delta\lambda\right) \le e^{-\frac{(\beta\lambda)^2}{2(\delta\lambda)^2}} = e^{-\frac{\beta^2}{2\delta^2}}.$$

Define $\tau = \inf\{u \ge 0\colon |M_u| > \lambda\}$.
Then

$$\begin{aligned}\mathsf{P}\left(M_\infty^* > \beta\lambda, [M]_\infty^{\frac{1}{2}} \le \delta\lambda\right) &= \mathsf{P}\left(M_\infty^* > \beta\lambda, [M]_\infty^{\frac{1}{2}} \le \delta\lambda, \tau < \infty\right)\\ &= \mathbb{E}\left[1_{\tau<\infty}\mathsf{P}\left(M_\infty^* > \beta\lambda, [M]_\infty^{\frac{1}{2}} \le \delta\lambda \,\middle|\, \mathcal{F}_\tau\right)\right]\end{aligned}$$

Define $N_t \equiv (M_{t+\tau} - M_\tau)^2 - ([M]_{t+\tau} - [M]_\tau)$.
Then $N_t \in \mathcal{F}_{t+\tau}$ is a continuous local martingale

$$\mathbb{E}\left[M_{t+\tau}^2 - [M]_{t+\tau} \mid \mathcal{F}_{s+\tau}\right] = M_{s+\tau}^2 - [M]_{s+\tau}.$$

$\implies N_0 = 0$ and on $\{M_\infty^* > \beta\lambda, [M]_\infty^{\frac{1}{2}} \le \delta\lambda\}$ we have $\exists\ \tilde{t} > 0$ s.t.

$$M_{\tilde{t}+\tau} \ge \beta\lambda \implies N_{\tilde{t}} = \left(M_{\tilde{t}+\tau} \pm \lambda\right)^2 - \left([M]_{\tilde{t}+\tau} - [M]_\tau\right) \ge [(\beta-1)\lambda]^2 - (\delta\lambda)^2.$$

Moreover $N_t \ge -\delta^2\lambda^2$, $\forall\ t \ge 0$.
Hence N_t reaches $[(\beta-1)\lambda]^2 - (\delta\lambda)^2$ before $(-\delta^2\lambda^2)$.
By (Gambler's Ruin $\mathsf{P}_k(\tau_0 < \tau_N) = \frac{N-k}{N}$)

$$\mathsf{P}(\tau_b < \tau_a) \le \frac{-a}{b-a}$$

$$\implies \mathsf{P}\left(M_\infty^* > \beta\lambda, [M]_\infty^{\frac{1}{2}} \le \delta\lambda \mid \mathcal{F}_\tau\right) \le \mathsf{P}\left(\tau_{[(\beta-1)\lambda]^2-(\delta\lambda)^2} < \tau_{-(\delta\lambda)^2}\right) \le \frac{(\delta\lambda)^2}{[(\beta-1)\lambda]^2} = \frac{\delta^2}{(\beta-1)^2}.$$

Here we take $0 < \delta < \beta - 1$. Therefore

$$\mathsf{P}\left(M_\infty^* > \beta\lambda, [M]_\infty^{\frac{1}{2}} \le \delta\lambda\right) \le \delta^2\frac{1}{(\beta-1)^2}\mathsf{P}(\tau < \infty) = \delta^2\frac{1}{(\beta-1)^2}\mathsf{P}\left(M_\infty^* > \lambda\right).$$

Similarly,

$$\mathsf{P}\left([M]_\infty^{\frac{1}{2}} > \beta\lambda, M_\infty^* \le \delta\lambda\right).$$

$$\tilde{N}_t = [M]_{t+\tilde{\tau}} - [M]_{\tilde{\tau}} - (M_{\tilde{\tau}+t} - M_{\tilde{\tau}})^2$$

with $\tilde{\tau} = \inf\{u \ge 0\colon [M]_u^{\frac{1}{2}} > \lambda\}$.

$$\implies \mathsf{P}\left([M]_\infty^{\frac{1}{2}} > \beta\lambda, M_\infty^* \le \delta\lambda, \tilde{\tau} < \infty\right) = \mathbb{E}\left[\mathsf{P}(\cdots \mid \mathcal{F}_{\tilde{\tau}})1_{\tilde{\tau}<\infty}\right].$$

Then $\tilde{N}_t$ reaches $(\beta\lambda)^2 - \lambda^2 - (\delta\lambda)^2$ before $-(\delta\lambda)^2$.

□

1.1 Semimartingale local time

For a Brownian motion $(B_t)_{t\ge 0}$. We consider

$$l_t^0 = \int_0^t \delta_0(B_s)\,\mathrm{d}s,$$

where δ_0 is the Dirac delta function such that $\forall$ function f,

$$\int f(x)\delta_0(x)\,\mathrm{d}x = f(0).$$

$$\mu([a,b]) = l_b^0 - l_a^0,$$

supp $(\mu) = \{t \ge 0\colon B_t = 0\}$, level set $= \{t \ge 0\colon B_t = x\}$, using local time

$$l_t^x = \int_0^t \delta_x(B_s)\,\mathrm{d}s.$$

Theorem 1.1.1 (Tanaka's formula)**.** Let X be a continuous semimartingale. Then $\exists$ continuous increasing adapted process $\{l_t, t \ge 0\}$

$$|X_t| - |X_0| = \int_0^t \operatorname{sgn}(X_s)\,\mathrm{d}X_s + l_t,$$

where

$$\operatorname{sgn}(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0. \end{cases}$$

Then $\{l_t, t \ge 0\}$ is called the local time of X at 0, and

$$\int_0^t 1_{\{X_s \ne 0\}}\,\mathrm{d}l_s = 0.$$

Remark.

(1) $f(x) = |x| \implies f'(x) = \operatorname{sgn}(x), \forall\, x \ne 0 \implies f''(x) = 2\delta(x)$. (distributional derivative)

(2) $|X_t - x| - |X_0 - x| = \int_0^t \operatorname{sgn}(X_0 - x)\,\mathrm{d}X_s + l_t^x$.

(3) Note that $|X_t| = X_t^+ + X_t^-$, $X_t = X_t^+ - X_t^-$

$$\implies 2X_t^+ = |X_t| + X_t = \int_0^t 1_{\{X_s>0\}}\,\mathrm{d}X_s - \int_0^t 1_{\{X_s\le 0\}}\,\mathrm{d}X_s + l_t + \int_0^t 1\,\mathrm{d}X_s + X_0 + |X_0|$$

$$\implies 2X_t^+ = 2\int_0^t 1_{\{X_s>0\}}\,\mathrm{d}X_s + l_t + 2X_0^+.$$

$$\implies X_t^+ - X_0^+ = \int_0^t 1_{\{X_s>0\}}\,\mathrm{d}X_s + \frac{1}{2}l_t.$$

Similarly, $2X_t^- = |X_t| - X_t$

$$\implies X_t^- - X_0^- = -\int_0^t 1_{\{X_s\le 0\}}\,\mathrm{d}X_s + \frac{1}{2}l_t.$$

Proof. Apply Itô's formula to the function $f(x) = |x|$.

$$\begin{aligned}\implies |X_t| &= |X_0| + \int_0^t f'(X_s)\,\mathrm{d}X_s + \frac{1}{2}\int_0^t f''(X_s)\,\mathrm{d}[X]_s \\ &= |X_0| + \int_0^t \operatorname{sgn}(X_s)\,\mathrm{d}X_s + \frac{1}{2}\underbrace{\int_0^t 2\delta(X_s)\,\mathrm{d}[X]_s}_{=l_t}.\end{aligned}$$

$\exists\ \varphi(x) \in C^\infty(\mathbb{R})$, define $\varphi_n(x) = \begin{cases} -1, & x \le 0 \\ \varphi(nx), & 0 < x < \frac{1}{n} \\ 1, & x \ge \frac{1}{n}, \end{cases}$ then $\varphi_n(x) \to \operatorname{sgn}(x)$, $\forall\ x \ne 0$.

Let

$$f_n(x) = \begin{cases} -x, & x \le 0 \\ \int_0^x \varphi_n(t)\,\mathrm{d}t, & x > 0. \end{cases}$$

$$\implies f_n'(x) = \varphi_n(x.)$$

Because $f(x) = |x| = \int_0^{|x|} 1\,\mathrm{d}t$.

$$\implies |f(x) - f_n(x)| \le \int_0^{|x|} |\varphi_n(t) - 1|\,\mathrm{d}t \le \int_0^{\frac{1}{n}} |\varphi_n(t) - 1|\,\mathrm{d}t \le 2\frac{1}{n}.$$

Apply Itô formula to f_n to get that

$$f_n(X_t) = f_n(X_0) + \int_0^t f_n'(X_s)\,\mathrm{d}X_s + \frac{1}{2}\int_0^t f_n''(X_s)\,\mathrm{d}[X]_s.$$

Note

$$\int_0^t f_n'(X_s)\,\mathrm{d}X_s = \int_0^t f_n'(X_s)\,\mathrm{d}M_s + \int_0^t f_n'(X_s)\,\mathrm{d}A_s,$$

and

$$\int_0^t f_n'(X_s)\,\mathrm{d}A_s = \int_0^t \varphi_n(X_s)\,\mathrm{d}A_s.$$

We have $\varphi_n(x) \uparrow \operatorname{sgn}(x)$, $\forall\, x$. By Monotone convergence,

$$\left|\int_0^t \varphi_n(X_s) - \operatorname{sgn}(X_s)\,\mathrm{d}A_s\right| \le \int_0^t |\varphi_n(X_s) - \operatorname{sgn}(X_s)|\,|\mathrm{d}A_s| \to 0,$$

which implies $\int_0^t \varphi_n(X_s)\,\mathrm{d}A_s \to \int_0^t \operatorname{sgn}(X_s)\,\mathrm{d}A_s$ uniform in t.

Next, $T_k = \inf\{t \ge 0\colon\ |M_t| \ge k\}$. Then M^{T_k} is bounded.

We may assume M is bounded. Then

$$\left\|\int_0^\infty \big(\operatorname{sgn}(X_s) - f_n'(X_s)\big)\,\mathrm{d}M_s\right\|_2^2 = \mathbb{E}\int_0^\infty \big(\operatorname{sgn}(X_s) - f_n'(X_s)\big)^2\,\mathrm{d}[M]_s \to 0.$$

So

$$\mathbb{E}\left[\sup_{t\le T}\left|\int_0^t \big(\operatorname{sgn}(X_s) - f_n'(X_s)\big)\,\mathrm{d}M_s\right|^2\right] \xrightarrow{n\to\infty} 0.$$

Hence

$$\sup_{t\ge 0}\int_0^t \big(\operatorname{sgn}(X_s) - f_n'(X_s)\big)\,\mathrm{d}M_s \xrightarrow{a.s.} 0.$$

$\implies \int_0^t f_n'(X_s)\,\mathrm{d}X_s \xrightarrow{a.s.} \int_0^t \operatorname{sgn}(X_s)\,\mathrm{d}X_s$, $f_n(X_t) \xrightarrow{a.s.} |X_t|$, $f_n(X_0) \xrightarrow{a.s.} |X_0|$.

$\implies C_t^n \triangleq \int_0^t \frac{1}{2} f_n''(X_s)\,\mathrm{d}[X_s]$, $C_t^n \xrightarrow{a.s.} l_t \triangleq |X_t| - |X_0| - \int_0^t \operatorname{sgn}(X_s)\,\mathrm{d}X_s$.

$\mathrm{d}C_s^n = \frac{1}{2} f_n''(X_s)\,\mathrm{d}[X]_s \implies \forall\, \delta > 0$, if $\frac{1}{n} < \delta$, then

$$|X_s| > \delta > \frac{1}{n} \implies f_n''(X_s) = 0.$$

$$\implies \int_0^t 1_{\{|X_s|>\delta\}}\,\mathrm{d}C_s^n = 0.$$

Hence

$$\int_0^t 1_{\{|X_s|>\delta\}}\,\mathrm{d}l_s \le \lim_{n\to\infty}\int_0^t 1_{\{|X_s|>\frac{\delta}{2}\}}\,\mathrm{d}C_s^n = 0.$$

$$\int_0^t \varphi_\delta(X_s)\,\mathrm{d}l_s = \lim_{n\to\infty}\int_0^t \varphi_\delta(X_s)\,\mathrm{d}C_s^n \le \liminf_{n\to\infty}\int_0^t 1_{\{|X_s|>\frac{\delta}{2}\}}\,\mathrm{d}C_s^n.$$

We prove $\int_0^t 1_{\{|X_s|>\delta\}}\,\mathrm{d}l_s = 0$, $\forall\, \delta > 0 \implies \int_0^t 1_{\{|X_s|\neq 0\}}\,\mathrm{d}l_s = 0$. □

Let X, Y be continuous semimartingale. Then

$$\int_0^t Y_t\,\mathrm{d}X_t = \text{Itô Integral}.$$

Stratonovich Integral

$$\int_0^t Y_t \partial X_t \triangleq \int_0^t Y_t \,\mathrm{d}X_t + \frac{1}{2}[X,Y]_t.$$

Recall that

$$\begin{aligned} X_tY_t - X_0Y_0 &= \int_0^t X_s \,\mathrm{d}Y_s + \int_0^t Y_s \,\mathrm{d}X_s + [X,Y]_t \\ &= \int_0^t X_s \,\mathrm{d}Y_s + \frac{1}{2}[X,Y]_t + \int_0^t Y_s \,\mathrm{d}X_s + \frac{1}{2}[X,Y]_t \\ &= \int_0^t X_s \partial Y_s + \int_0^t Y_s \partial X_s. \end{aligned}$$

Theorem 1.1.2. $f(X_t) = f(X_0) + \int_0^t f'(X_s)\partial X_s$.

Proof. By Itô formula,

$$f(X_t) = f(X_0) + \int_0^t f'(X_s)\,\mathrm{d}X_s + \frac{1}{2}\int_0^t f''(X_s)\,\mathrm{d}[X_s].$$

By definition,

$$\int_0^t f'(X_s)\partial X_s = \int_0^t f'(X_s)\,\mathrm{d}X_s + \frac{1}{2}\left[f'(X), X\right]_t.$$

$$\mathrm{d}f(X_t) = f'(X_t)\,\mathrm{d}X_t + \frac{1}{2}f''(X_t)\,\mathrm{d}[X]_t$$

$$\implies \mathrm{d}f'(X_t) = f''(X_t)\,\mathrm{d}X_t + \frac{1}{2}f'''(X_t)\,\mathrm{d}[X]_t.$$

So

$$f'(X_t) = f'(X_0) + \int_0^t f''(X_s)\,\mathrm{d}X_s + \frac{1}{2}\int_0^t f'''(X_s)\,\mathrm{d}[X]_s.$$

$$\implies \left[f'(X), X\right]_t = \left[\int_0^t f''(X_s)\,\mathrm{d}X_s, X_t\right] = \int_0^t f''(X_s)\,\mathrm{d}[X]_s.$$

($[H\cdot M, K\cdot N]_t = \int_0^t HK\,\mathrm{d}[M,N]_s$).

So

$$f(X_t) = f(X_0) + \int_0^t f'(X_s)\,\mathrm{d}X_s + \frac{1}{2}\left[f'(X), X\right]_t = f(x_0) + \int_0^t f'(X_s)\partial X_s.$$

□

Theorem 1.1.3 (Riemam-Sum Approximation)**.** Let C be a locally bounded adapted L-process. Let X be a continuous semimartingale. Then

$$\sum_{i=1}^{\infty} c(t_{i-1}^n)(X_{t_i^n} - X_{t_{i-1}^n}) \to \int_0^t c(s)\,\mathrm{d}X_s.$$

with $t_i^n = t \wedge \frac{i}{2^n}$. The convergence is uniform in probability on compacts, i.e. $\forall\ K > 0$,

$$\sup_{t \le K} |I^n(t) - I(t)| \xrightarrow{P} 0.$$

Proof. Let $c^n(t) = c(t_{i-1}^n)$ for $t \in (t_{i-1}^n, t_i^n]$. Then $c^n(t) \to c(t)$ for all $t \ge 0$.

Also,

$$\begin{aligned} I^n(t) &= \sum_{i=1}^{\infty} c(t_{i-1}^n)(X(t_i^n) - X(t_{i-1}^n)) \\ &= \int_0^t c^n(s)\,\mathrm{d}X_s \\ &= (c^n, X)_t = \int_0^t c^n(s)\,\mathrm{d}M_s + \int_0^t c^n(s)\,\mathrm{d}A_s. \end{aligned}$$

$\forall\ t \le k$, $|c^n(t)| \le c_k$

$$\implies \int_0^t c^n(s)\,\mathrm{d}A_s \to \int_0^t c(s)\,\mathrm{d}A_s.$$

By localization, we may assume M is bounded. If c is bounded and $M \in \mathcal{M}_0^2$, then

$$\int_0^t c^n(s)\,\mathrm{d}M_s \to \int_0^t c(s)\,\mathrm{d}M_s.$$

$$\mathbb{E}\left[\sup_{t \le k}\left|\int_0^t (c^n(s) - c(s))\,\mathrm{d}M_s\right|^2\right] \le c_k \mathbb{E}\int_0^k (c^n(s) - c(s))^2\,\mathrm{d}[M]_s \to 0.$$

□

1.1.1 Stratonovich Calculus

Let X, Y be continuous semimartingale. Define

$$\begin{aligned} S^n(t) &\equiv \sum_{i=1}^{n} \frac{1}{2}\left(Y(t_i^n) + Y(t_{i-1}^n)\right)\left(X(t_i^n) - X(t_{i-1}^n)\right) \\ S(t) &= \int_0^t Y_s \partial X_s. \end{aligned}$$

Then

$$\sup_{t \le k} |S^n(t) - S(t)| \xrightarrow{P} 0.$$

Proof.

$$\begin{aligned} S^n(t) &= \sum_{i=1}^{\infty} \left[Y(t_{i-1}^n) + \frac{1}{2}(Y(t_i^n) - Y(t_{i-1}^n)) \right] (X(t_i^n) - X(t_{i-1}^n)) \\ &= \sum_{i=1}^{\infty} Y(t_{i-1}^n)(X(t_i^n) - X(t_{i-1}^n)) + \frac{1}{2} \sum_{i=1}^{\infty} \Delta Y(t_i^n) \Delta X(t_i^n). \end{aligned}$$

It remains to prove

$$\begin{aligned} &\sum_{i=1}^{\infty} \Delta Y(t_i^n) \Delta X(t_i^n) \to [X,Y]_t = \frac{1}{4}([X+Y]_t - [X-Y]_t) \\ &\Leftrightarrow \sum_{i=1}^{\infty} \Delta X(t_i^n)^2 \to [X]_t \\ &\Leftrightarrow \sum_{i=1}^{\infty} (\Delta M(t_i^n) + \Delta A(t_i^n))^2 \to [X]_t = [M]_t. \end{aligned}$$

We have $\sum_{i=1}^{\infty} (\Delta M(t_i^n))^2 \to [M]_t$,

$$\left| \sum_{i=1}^{\infty} \Delta M(t_i^n) \Delta A(t_i^n) \right| \le \sup_{i \ge 1} |\Delta M(t_i^n)| \sum_{i=1}^{\infty} |\Delta A(t_i^n)| \le \sup_{\substack{t \le k \\ i \ge 1}} |\Delta M(t_i^n)| \operatorname{Var}_A(t) \xrightarrow{a.s.} 0.$$

□

1.2 Stochastic Differenctial Equation

$$\begin{aligned} &\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + \sigma(t, X_t)\,\mathrm{d}W_t, \\ &X_t = X_0 + \int_0^t b(s, X_s)\,\mathrm{d}s + \int_0^t \sigma(s, X_s)\,\mathrm{d}W_s. \end{aligned}$$

d-dimension SDE ...

Definition 1.2.1 (Strong solution)**.** A strong solution of the SDE, on a given probability space $(\Omega, \mathcal{F}, \mathsf{P})$ and a given B.M. $(W_t, \mathcal{F}_t^W)_{t \ge 0}$, with initial condition ξ is a process $\{X_t, t \ge 0\}$ with continuous sample path and

(i) $X \in \mathcal{F}_t$ with $(\mathcal{F}_t)_{t \ge 0}$ being the augmented filtration of the B.M. $(\mathcal{F}_t^W)_{t \ge 0}$.

(ii) $\mathsf{P}(X_0 = \xi) = 1$.

(iii)

$$\mathsf{P}\left(\int_0^t |b_i(s, X_s)| + \sigma_{ij}^2(s, X_s)\,\mathrm{d}s < \infty \right) = 1$$

for all i, j, t.

(iv) $X_t = X_0 + \int_0^t b(s, X_s)\,\mathrm{d}s + \int_0^t \sigma(s, X_s)\,\mathrm{d}W_s$.

Definition 1.2.2 (Strong Uniqueness)**.** Given $b(t,x)$ and $\sigma(t,x)$. Suppose $\forall$ B.M. $\{W_t\}_{t\geq 0}$ on some $(\Omega, \mathcal{F}, \mathsf{P})$ and ξ is the initial condition, and $\{\mathcal{F}_t\}$ is the augmented filtration, if X and $\tilde{X}$ are two strong solutions with respect to the sam W, then

$$\mathsf{P}\left(X_t = \tilde{X}_t, \forall\, t \geq 0\right) = 1.$$

Example . (i) $\mathrm{d}X_t = \mathrm{d}W_t \implies X_t = W_t + \xi_0$.

(ii) $\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + \mathrm{d}W_t$ where $b : [0,\infty) \times \mathbb{R} \to \mathbb{R}$ is bounded Borel-mble and $\forall\, x \leq y$, we have $b(t,x) \geq b(t,y)$, $\forall\, t \geq 0$. Fot this SDE, strong uniqueness holds.

Proof. Eg.(ii) $\forall$ B.M. on $(\Omega, \mathcal{F}, \mathsf{P})$, $\forall\, \xi$. If $X^{(1)}$ & $X^{(2)}$ are strong solutions, then

$$X_t^{(j)} = \xi + \int_0^t b(s, X_s^{(j)})\,\mathrm{d}s + W_t, \quad \forall\, j = 1, 2.$$

$$X_t^{(1)} - X_t^{(2)} = \int_0^t \left[b(s, X_s^{(1)}) - b(s, X_s^{(2)})\right] \mathrm{d}s.$$

$$\implies \mathrm{d}\Delta_t = \left[b(t, X_t^{(1)}) - b(t, X_t^{(2)})\right] \mathrm{d}t.$$

By Itô's formula,

$$\Delta_t^2 = \int_0^t 2\Delta_s\,\mathrm{d}\Delta_s + \frac{1}{2}\int_0^t 2\,\mathrm{d}[\Delta]_t = 2\int_0^t \left(X_s^{(1)} - X_s^{(2)}\right)\left[b(s, X_s^{(1)}) - b(s, X_s^{(2)})\right] \mathrm{d}s.$$

If $x \leq y$, then $b(s,x) - b(s,y) \geq 0$.
If $x > y$, then $b(s,x) - b(s,y) \leq 0$.
$\implies (x-y)(b(s,x) - b(s,y)) \leq 0$. So

$$\Delta_t^2 = 2\int_0^t \left(X_s^{(1)} - X_s^{(2)}\right)\left[b(s, X_s^{(1)}) - b(s, X_s^{(2)})\right] \mathrm{d}s \leq 0, \quad \text{a.s.}$$

$\implies \mathsf{P}\left(\Delta_t = 0, \forall\, t \geq 0\right) = 1.$
$\implies$ strong uniqueness. □

Theorem 1.2.1 (Itô Uniqueness Theorem)**.** Suppose $b(t,x)$ and $\sigma(t,x)$ are locally Lipschitz in the space variable: $\forall\, n \geq 1$, $\exists\, K_n > 0$ s.t. $\forall\, t \geq 0$, $\|x\|, \|y\| \leq n$, we have

$$\|b(t,x) - b(t,y)\| + \|\sigma(t,x) - \sigma(t,y)\| \leq K_n \|x - y\|.$$

Then strong uniqueness holds.

Lemma 1.2.1 (Gronwall's Inequality)**.** Suppose a continuous function $g(t)$ satisfies $0 \leq g(t) \leq \alpha(t) + \beta \int_0^t g(s)\,\mathrm{d}s$ for all $t \in [0,T]$ with $\beta \geq 0$ and $\alpha : [0,T] \to \mathbb{R}$ integrable. Then

$$g(t) \leq \alpha(t) + \beta \int_0^t \alpha(s) e^{\beta(t-s)}\,\mathrm{d}s, \quad \forall\, t \leq T.$$

Proof. $I(t) := \int_0^t g(s)\,\mathrm{d}s$.

$\implies I'(t) = g(t) \le \alpha(t) + \beta I(t)$.

$\implies I'(t) - \beta I(t) - \alpha(t) \le 0$.

Notice that

$$\left[e^{-\beta t} I(t)\right]' = \left(I'(t) - \beta I(t)\right) e^{-\beta t} \le \alpha(t) e^{-\beta t}.$$

$e^{-\beta t} I(t) - 0 = \int_0^t \left[e^{-\beta s} I(s)\right]' \mathrm{d}s \le \int_0^t \alpha(s) e^{-\beta s}\,\mathrm{d}s$.

$\implies I(t) \le \int_0^t \alpha(s) e^{\beta(t-s)}\,\mathrm{d}s$.

$\implies g(t) \le \alpha(t) + \beta I(t) = \alpha(t) + \beta \int_0^t \alpha(s) e^{\beta(t-s)}\,\mathrm{d}s$. □

Proof. of Itô Uniqueness Theorem.

Given W and $(\Omega, \mathcal{F}, \mathsf{P})$ and ξ. If X and $\tilde{X}$ are two strong solutions with respect to the same W. Define

$$\tau_n = \inf\{t \ge 0\colon\ \|X_t\| \ge n\}$$
$$\tilde{\tau}_n = \inf\{t \ge 0\colon\ \left\|\tilde{X}_t\right\| \ge n\}.$$

Set $\tau_n \wedge \tilde{\tau}_n$. Then $\lim_{n\to\infty} S_n = \infty$, P-a.s.

Note that

$$X_{t\wedge S_n} - \tilde{X}_{t\wedge S_n} = \int_0^{t\wedge S_n} \left[b(s, X_s) - b(s, \tilde{X}_s)\right] \mathrm{d}s + \int_0^{t\wedge S_n} \left[\sigma(s, X_s) - \sigma(s, \tilde{X}_s)\right]$$

$$\implies \mathbb{E}\left\|X_{t\wedge S_n} - \tilde{X}_{t\wedge S_n}\right\|^2 \le 4\mathbb{E}\left[\int_0^{t\wedge S_n} (b - b)\,\mathrm{d}s\right]^2 + 4\mathbb{E}\left[\int_0^{t\wedge S_n} (\sigma - \sigma)\,\mathrm{d}W_s\right]^2$$

.

$(X_t^1, X_t^2, \ldots, X_t^d) = \left(\int_0^t b^1\,\mathrm{d}s + \int_0^t \sigma^1\,\mathrm{d}W_s, \ldots, \int_0^t b^d\,\mathrm{d}s + \int_0^t \sigma^d\,\mathrm{d}W_s\right)$.

$$\begin{aligned}
\implies \|X\|^2 &= \sum_{i=1}^d (X_t^j)^2 \\
&= \sum_{i=1}^d \left(\int_0^t b^j\,\mathrm{d}s + \int_0^t \sigma^j\,\mathrm{d}W_s\right)^2 \\
&\le 2\sum_{i=1}^d \left(\int_0^t b^j\,\mathrm{d}s\right)^2 + \left(\int_0^t \sigma^j\,\mathrm{d}W_s\right)^2
\end{aligned}$$

$$
\begin{aligned}
\mathbb{E}\left\|X_{t\wedge S_n}-\tilde{X}_{t\wedge S_n}\right\|^2 \leq & 2\sum_{i=1}^{d}\mathbb{E}\left(\int_0^{t\wedge S_n}\left(b_i(s,X_s)-b(s,\tilde{X}_s)\right)\mathrm{d}s\right)^2 \\
& +2\sum_{i=1}^{d}\mathbb{E}\left(\sum_{j=1}^{r}\int_0^{t\wedge S_n}\left(\sigma_{ij}(s,X_s)-\sigma_{ij}(s,\tilde{X}_s)\right)\mathrm{d}W_s^{(j)}\right)^2 \\
\leq & 2\sum_{i=1}^{d}\mathbb{E}\left[t\cdot\int_0^{t\wedge S_n}\left(b_i(s,X_s)-b_i(s,\tilde{X}_s)\right)^2\mathrm{d}s\right] \\
& +2\sum_{i=1}^{d}\sum_{j=1}^{r}\mathbb{E}\left[\int_0^{t\wedge S_n}(\sigma_{ij}-\sigma_{ij})\,\mathrm{d}W_s^{(j)}\right]^2
\end{aligned}
$$

And $\mathbb{E}\left[\int_0^{t\wedge S_n}(\sigma_{ij}-\sigma_{ij})\,\mathrm{d}W_s^{(j)}\right]^2=\mathbb{E}\left[\int_0^{t\wedge S_n}(\sigma_{ij}-\sigma_{ij})^2\,\mathrm{d}s\right]$.

We conclude that

$$
\begin{aligned}
\mathbb{E}\left\|X_{t\wedge S_n}-\tilde{X}_{t\wedge S_n}\right\|^2 \leq & 2t\mathbb{E}\int_0^{t\wedge S_n}\left\|b(s,X_s)-b(s,\tilde{X}_s)\right\|^2\mathrm{d}s+2\mathbb{E}\int_0^{t\wedge S_n}\left\|\sigma(s,X_s)-\sigma(s,\tilde{X}_s)\right\|^2\mathrm{d}s \\
\leq & 2T\mathbb{E}\int_0^{t\wedge S_n}K_n^2\left\|X_s-\tilde{X}_s\right\|^2\mathrm{d}s+2\mathbb{E}\int_0^{t\wedge S_n}K_n^2\left\|X_s-\tilde{X}_s\right\|^2\mathrm{d}s \\
\leq & 2(T+1)K_n^2\int_0^t\mathbb{E}\left\|X_{s\wedge S_n}-\tilde{X}_{s\wedge S_n}\right\|^2\mathrm{d}s,
\end{aligned}
$$

since

$$
\|\sigma\|^2=\sum_{i=1}^{d}\sum_{j=1}^{r}\sigma_{ij}^2.
$$

If we let $g(t)=\triangleq\mathbb{E}\left\|X_{t\wedge S_n}-\tilde{X}_{t\wedge S_n}\right\|^2$, then $g(t)\leq 2(T+1)K_n^2\int_0^t g(s)\,\mathrm{d}s$.

By Gronwall's inequality, we get $g(t)\leq 0$.

$\implies X_{t\wedge S_n}=\tilde{X}_{t\wedge S_n}$ a,s, $\forall\ t\geq 0$.

$\implies X_{t\wedge S_n}=\tilde{X}_{t\wedge S_n}$, $\forall\ t\geq 0$, a.s.

Let $S_n\uparrow\infty$ to conclude $X_t=\tilde{X}_t$, $\forall\ t\geq 0$, a.s. □

Theorem 1.2.2 (Existence)**.** Suppose $\exists\ K>0$ s.t. $\forall\ t\geq 0$, $\forall\ x,y\in\mathbb{R}^d$,

$$
\begin{cases}
\|b(t,x)-b(t,y)\|+\|\sigma(t,x)-\sigma(t,y)\|\leq K\|x-y\| \\
\|b(t,x)\|+\|\sigma(t,x)\|\leq K(1+\|x\|).
\end{cases}
$$

On $(\Omega,\mathcal{F},\mathsf{P})$, let ξ be R.V. and r-dimensional $B.M.$ $W=\{W_t\}_{t\geq 0}$ independent of ξ, and $\mathbb{E}\|\xi\|^2<\infty$. Let $(\mathcal{F}_t)_{t\geq 0}$ be the augmented filtration. Then $\exists$ continuous adapted process $X=\{X_t\}_{t\geq 0}$ which is a strong solution with respect to W and $X_0=\xi$ a.s.

Moreover, $\forall\ T>0$, $\exists\ C=C(K,T)>0$ s.t. $\forall\ 0\leq t\leq T$,

$$
\mathbb{E}\|X_t\|^2\leq C(1+\mathbb{E}\|\xi\|^2)e^{Ct}.
$$

Proof. Define $X_t^{(0)} = \xi$, $\forall\, t \geq 0$.

For any $k \geq 1$, define

$$X_t^{(k)} \triangleq \xi + \int_0^t b(s, X_s^{(k-1)})\,\mathrm{d}s + \int_0^t \sigma(s, X_s^{(k-1)})\,\mathrm{d}W_s.$$

Then $\forall\, T > 0$, $\exists\, C = C(K,T) > 0$ s.t.

$$\mathbb{E}\left\|X_t^{(k)}\right\|^2 \leq C(1 + \mathbb{E}\,\|\xi\|^2)e^{Ct}, \quad \forall\, t \leq T, \forall\, k \geq 0.$$

(Exercise)

For any $k \geq 0$,

$$X_t^{(k+1)} - X_t^{(k)} = \int_0^t \left(b(s, X_s^{(k)}) - b(s, X_s^{(k-1)})\right)\mathrm{d}s + \int_0^t \left(\sigma(s, X_s^{(k)}) - \sigma(s, X_s^{(k-1)})\right)\mathrm{d}W_s.$$

$$\begin{aligned}
\mathbb{E}\left[\sup_{s\leq t}\|B_s\|^2\right] &= \mathbb{E}\left\|\sup_{s\leq t}\int_0^s \left(b(u, X_u^{(k)}) - b(u, X_u^{(k-1)})\right)\mathrm{d}u\right\|^2 \\
&\leq \mathbb{E}\left[t \cdot \int_0^t \left\|b(u, X_u^{(k)}) - b(u, X_u^{(k-1)})\right\|^2 \mathrm{d}u\right] \\
&\leq K^2 + \int_0^t \mathbb{E}\left\|X_u^{(k)} - X_u^{(k-1)}\right\|^2 \mathrm{d}u.
\end{aligned}$$

$M_t = \int_0^t \sigma(s, X_s^{(k)}) - \sigma(s, X_s^{(k-1)})\,\mathrm{d}W_s$.

$[M]_t = \int_0^t \left\|\sigma(s, X_s^{(k)}) - \sigma(s, X_s^{(k-1)})\right\|^2 \mathrm{d}s$.

$\implies \mathbb{E}[M]_t \leq \int_0^t K^2 \left(1 + \left\|X_s^{(k)}\right\| + \left\|X_s^{(k-1)}\right\|\right)^2 \mathrm{d}s < \infty$ by $\mathbb{E}\left\|X_t^{(k)}\right\|^2 \leq Ce^{Ct}$.

$$\begin{aligned}
\mathbb{E}\left[\sup_{s\leq t}\|M_s\|^2\right] &\leq C\mathbb{E}[M_t] \\
&= C\mathbb{E}\int_0^t \left\|\sigma(s, X_s^{(k)}) - \sigma(s, X_s^{k-1})\right\|^2 \mathrm{d}s \\
&\leq C\mathbb{E}\int_0^t K\left\|X_s^{(k)} - X_s^{(k-1)}\right\|^2 \mathrm{d}s.
\end{aligned}$$

Combine the two inequalities to get

$$\begin{aligned}
\mathbb{E}\left[\max_{s\leq t}\left\|X_s^{(k+1)} - X_s^{(k)}\right\|^2\right] &\leq 2\mathbb{E}\left[\max_{s\leq t}\|B_s\|^2\right] + 2\mathbb{E}\left[\max_{s\leq t}\|M_s\|^2\right] \\
&\leq C \cdot 4K^2(1+T)\int_0^t \mathbb{E}\left\|X_s^{(k)} - X_s^{(k-1)}\right\|^2 \mathrm{d}s \\
&\leq L\int_0^t \mathbb{E}\left(\left\|X_u^{(k)} - X_u^{(k-1)}\right\|^2\right)\mathrm{d}s.
\end{aligned}$$

When $k = 1$, we have

$$\mathbb{E}\left(\left\|X_s^{(1)} - X_s^{(0)}\right\|^2\right) \leq 2\mathbb{E}\left\|X_t^{(1)}\right\|^2 + 2\mathbb{E}\left\|X_t^{(0)}\right\|^2 \leq C^*$$

for all $0 \leq t \leq T$.

Hence

$$\mathbb{E}\left(\max_{0\leq s\leq t}\left\|X_s^{(2)} - X_s^{(1)}\right\|^2\right) \leq L\int_0^t C^* \,\mathrm{d}s = C^*(Lt).$$

$$\implies \mathbb{E}\left(\max_{0\leq s\leq t}\left\|X_s^{(3)} - X_s^{(2)}\right\|^2\right) \leq L\int_0^t C^*(Ls)\,\mathrm{d}s = C^*\frac{1}{2}(Lt)^2.$$

By induction, we can prove $\forall\ k \geq 1$,

$$\mathbb{E}\left(\max_{s\leq t}\left\|X_s^{(k+1)} - X_s^{(k)}\right\|^2\right) \leq C^*\frac{1}{k!}(Lt)^k.$$

By chebyshev's Inequality,

$$\mathsf{P}\left(\max_{0\leq t\leq T}\left\|X_t^{(k+1)} - X_t^{(k)}\right\| > \frac{1}{2^{k+1}}\right) \leq 4^{k+1}C^*\frac{1}{k!}(LT)^k.$$

$\sum_{k=1}^{\infty}\mathsf{P}(\max\|\ldots\| > \frac{1}{2^{k+1}}) < \infty$.

$\implies$ B.C. Lemma gives

$$\mathsf{P}\left(\max_{t\leq T}\left\|X_t^{(k+1)} - X_t^{(k)}\right\| > \frac{1}{2^{k+1}}\ \text{i.o.}\right) = 0.$$

$\implies$ P-a.s. $\exists\ N(\omega) < \infty$ s.t. $\forall\ k \geq N$,

$$\max_{t\leq T}\left\|X_t^{(k+1)} - X_t^{(k)}\right\| \leq \frac{1}{2^{k+1}}.$$

$\implies$ $\forall\ k \geq N$, $\forall\ m \geq 1$, we have

$$\max_{t\leq T}\left\|X_t^{(k+m)} - X_t^{(k)}\right\| \leq \sum_{j=k}^{k+m-1}\frac{1}{2^{j+1}} \leq \frac{1}{2^k}.$$

Hence $\left\{X_t^{(k)}(\omega), 0 \leq t \leq T\right\}$ is a Cauchy sequence in $\|\ldots\|_{sup}$.

Hence $\exists$ limit $\{X_t(\omega), 0 \leq t \leq T\}$ s.t.

$$\lim_{k\to\infty}\max_{t\leq T}\left\|X_t^{(k)} - X_t\right\| = 0.$$

Since $T > 0$ is arbitrary, $\exists\ X = \{X_t, 0 \leq t < \infty\}$ s.t. $\forall\ T > 0$, $\sup_{t\leq T}\left\|X_t^{(k)} - X_t\right\| \to 0$ as $k \to \infty$.

$$\begin{aligned}\mathbb{E}\left\|X_t\right\|^2 &= \mathbb{E}\lim_{k\to\infty}\left\|X_t^{(k)}\right\|^2 \\ &\le \lim_{k\to\infty}\mathbb{E}\left\|X_t^{(k)}\right\|^2 \le C(1+\mathbb{E}\left\|\xi\right\|^2)e^{Ct}.\end{aligned}$$

$$\implies \mathsf{P}\left(\int_0^t |b(s,X_s)|\,\mathrm{d}s + \int_0^t \sigma_{ij}^2(s,X_s)\,\mathrm{d}s < \infty\right) = 1.$$

$$X_t^{(k+1)} = \xi + \int_0^t b(s,X_s^{(k)})\,\mathrm{d}s + \int_0^t \sigma(s,X_s^{(k)})\,\mathrm{d}W_s.$$

$$X_t = \xi + \int_0^t b(s,X_s)\,\mathrm{d}s + \int_0^t \sigma(s,X_s)\,\mathrm{d}W_s.$$

Need to prove $X_t^{(k+1)} \xrightarrow{k\to\infty} X_t$ a.s. □

Theorem 1.2.3 (Yamada-Watanabe)**.** Let $\mathrm{d}X_t = \sigma(X_t)\,\mathrm{d}W_t + b(X_t)\,\mathrm{d}t$, where

$$\sigma(x) = \begin{pmatrix}\sigma_1(x_1) & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_n(x_)\end{pmatrix}$$

$$b(x) = (b^1(x), b^2(x), \ldots, b^n(x))$$

such that

(i) $\exists$ positive increasing function $\rho(u)$, $u \in (0,\infty)$ s.t.

$$|\sigma_i(\xi) - \sigma_i(\eta)| \le \rho\left(|\xi-\eta|\right)$$

for all $\xi,\eta \in \mathbb{R}^1$, $i = 1,\ldots,n$ and

$$\int_{0^+} \rho^{-2}(u)\,\mathrm{d}u = \infty.$$

i.e. $\forall\ \varepsilon > 0$, $\int_0^\varepsilon \rho^{-2}(u)\,\mathrm{d}u = \infty$.

(ii) $\exists$ positive increasing concave function $k(u)$, $u \in (0,\infty)$ s.t.

$$|b_i(x) - b_j(y)| \le k\left(|x-y|\right)$$

for all $x,y \in \mathbb{R}^n$, $i = 1,\ldots,n$ and $\int_{0^+} k^{-1}(u)\,\mathrm{d}u = \infty$.

Then strong uniqueness holds.

Remark. If $\rho(u) = u^\alpha$, then strong uniqueness holds if $\alpha \ge \frac{1}{2}$; fails if $0 < \alpha < \frac{1}{2}$.

Definition 1.2.3 (defn example)**.** Example

Theorem 1.2.4 (thm example)**.** Example

Proof. proof □

Lemma 1.2.2 (lemma example)**.** Example

Proof for Lemma.

■. proof for lemma example □

Proposition 1.2.1. pro example

Proof for Proposition.

■. Example □

Claim 1.2.1. claim example Example

Proof for Claim.

■. proof for claim example □

Corollary 1.2.1 (cor example)**.** example

Proof for Corollary.

■. proof for corollary example □

Fact 1.2.1. Example fact

Remark. Example remark

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